

ONE RENEGADE CELL: A REVIEW ON THE STRENGTHS AND LIMITS OF WEINBERG'S VIEWS

Atin Thakur,

IISER Pune, India.

Robert Weinberg's *One Renegade Cell* (1998) explains how a single cell turns cancerous and starts dividing. This cell bypasses several checkpoints and evades apoptosis. Gradually, it grows uncontrollably and may form a tumour, unnoticeably. Weinberg explains cancer as a step-by-step process. Carcinogens-cancer causing agents-include radiation, chemicals such as asbestos and aflatoxins and certain viruses. These agents can damage the DNA in cells. According to him, cancer can initiate because of two major genetic mechanisms; one in which normal genes become oncogenes and another in which the Tumour Suppressor Genes, which normally prevent cancer, become inactive. He highlights the "two-hit hypothesis proposed by Knudson," especially for TSGs like the *RBI* gene (Retinoblastoma), which means both copies of the gene must be damaged for cancer. He also gives an example of a small change in DNA like point mutation i.e. GGC becoming GTC in bladder cancer.

The book explains several genetic events that are required for a cell to become cancerous. Simply put, the internal checks of cells start to fail. For instance, Colorectal cancer, the sequence involves activation of oncogenes like *Ras* (Rat Sarcoma) and inactivation of TSGs e.g. *APC* (Adenomatous Polyposis Coli gene), *DCC* (Deleted in Colorectal Cancer gene), and *p53* (Tumor protein 53). This allows multistep cancer progression. He briefs about non-mutagenic factors, such as hormones that result in increased proliferation. Genes like *RBI* and *p53* help regulate the cell cycle, and their loss leads to continuous cell proliferation, defective DNA repair, and resistance to apoptosis. This results in clonal selection, where the most abnormal cells survive under stress conditions. Telomerase activation and angiogenesis (via VEGF; Vascular Endothelial Growth Factor) are mentioned briefly. These most probable genetic and epigenetic changes help cancer cells become more independent and able to thrive, influencing gene expression and cellular processes. However, I felt the book could have a detailed explanation of DNA-protein interactions in damage and repair processes.

The sections on EGFR inhibitors and BCR-ABL proteins act as a direction towards advancements in future technologies. Ultimately, the book provides foundational knowledge and blueprints of cancer progression. This understanding has led to the development of highly discussed cutting-edge modern techniques like CRISPR widely developed after 2012 and progressive developments in viral vector therapy; most recent in 21st century : vectors that evade or reduce immune detection and cargos deliver their genetic cargo into specific organs like liver.

The strength of this book includes explanation of experimental concepts including real case studies like the Ames test, and cancer progression in some types like colorectal cancer. Weaknesses include a lack of discussion on cell-level regulation, immune evasion and crucial regulators of epigenetic effects on DNA methylation and histone modifications. The claim, "most TSGs (except *RBI*) are not inherited" appears to be partially incorrect and misleading as there are several well documented cases of germline inheritance like *APC* & *TP53*.

I would recommend the book as a basic reference on cancer for students like me as it provides an idea of how normal cells can turn cancerous, primarily because of mutations. Despite the limited scope, the book tries to conclude all the aspects for anyone entering or interested in the field of cancer biology.